

Student: Jonatan Rabjerg
Date: 1/15/24

Instructor: Andreas Basse-O'Connor
Course: Calculus Beta

Assignment: MyLab Math - Uge 4

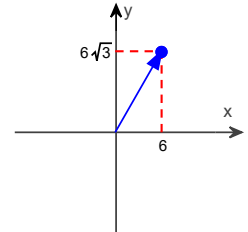
Express the complex number in the form $re^{i\theta}$ with $r \geq 0$ and $-\pi < \theta \leq \pi$, and draw a diagram in the complex plane for it.

$$(6 + \sqrt{-108})^2$$

Start by rewriting the number in parentheses in the form $a + ib$.

$$(6 + \sqrt{-108})^2 = (6 + 6i\sqrt{3})^2$$

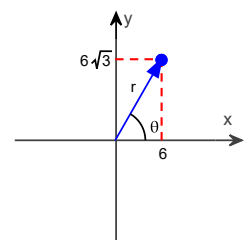
The next step is to express the number $6 + 6i\sqrt{3}$ in the form $re^{i\theta}$. To do this, first plot it in the complex plane. This is shown here.



Use this figure to find the values of r and θ , with $r \geq 0$ and $-\pi < \theta \leq \pi$, for the number $6 + 6i\sqrt{3}$.

$$r = 12$$

$$\theta = \frac{\pi}{3}$$



Rewrite the expression from earlier using these values of r and θ .

$$(6 + 6i\sqrt{3})^2 = \left(12e^{\frac{i\pi}{3}}\right)^2$$

Simplify this result.

$$\left(12e^{\frac{i\pi}{3}}\right)^2 = 144e^{\frac{2i\pi}{3}}$$

This is in the form $re^{i\theta}$ with $r \geq 0$ and $-\pi < \theta \leq \pi$ because $r = 144$ is positive and $\theta = \frac{2\pi}{3}$ is between $-\pi$ and π .

$$\text{Thus, } (6 + \sqrt{-108})^2 = 144e^{\frac{2i\pi}{3}}.$$

A complex number $z = x + iy$ can be expressed as a vector from the origin to the point $P(x,y)$. If z is expressed in the form $re^{i\theta}$ with $r \geq 0$, then r is the length of the vector and θ is the angle the vector makes with the positive x -axis.

$$\text{In the case of } 144e^{\frac{2i\pi}{3}}, r = 144 \text{ and } \theta = \frac{2\pi}{3}.$$

Use this to draw a diagram in the complex plane for $144e^{\frac{2i\pi}{3}}$. This is shown here.

